

Material Minder: A Voice-Enabled Mobile Application for Managing Construction Materials

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May 16, 2025

Contents

1	Introduction	3
1.1	Background on Challenges in Construction Material Management	3
1.2	Introduction to Material Minder	3
2	Theoretical Background	4
2.1	Human-Computer Interaction (HCI) and Universal Design	4
2.2	Speech and Language Processing	6
2.3	Assistive Technologies	7
3	Application Description	8
3.1	Project Management:	8
3.2	Material Management:	8
3.3	Voice-Enabled Input:	9
3.4	Cost Estimation:	9
3.5	Non-Functional Requirements	9
3.6	System Specification for Material Minder	10
4	Functional Specifications	10
4.1	Project Management Specifications:	10
4.2	Material Management Specifications:	11
4.3	Voice-Enabled Input Specifications:	12
4.4	Cost Estimation Specifications:	13
4.5	Non-Functional Specifications	13
5	Development and Implementation	14
6	Results and Evaluation	14
7	Discussion	14
8	Conclusion	14

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Executive Summary

The Material Minder mobile application is designed to assist handymen, particularly elderly individuals or those with disabilities, in managing construction materials efficiently. The app provides an intuitive and accessible platform for organizing materials through a CRUD-based system, allowing users to create, update, and manage lists of required materials for various projects. Additionally, it includes a customizable database of construction materials sourced from Romanias largest suppliers, while also enabling users to remove unnecessary materials or add custom items that may not be readily available in stores.

The key innovation of Material Minder is its voice-enabled input system, which enhances accessibility and ease of use. Instead of manually entering data, users can simply dictate the materials they need, and the application will process the input, automatically generating a structured materials list. This feature significantly reduces the effort required for material management, making the app especially useful for those who may have difficulty using traditional mobile interfaces. By integrating voice recognition technology with a user-friendly material tracking system, this project aims to improve efficiency and accessibility in construction-related tasks, ultimately enhancing productivity for its target users.

1 Introduction

1.1 Background on Challenges in Construction Material Management

The effective management of construction materials is a multifaceted process involving meticulous planning, precise purchasing, continuous tracking, organized storage, and diligent minimization of waste. These activities, while seemingly straightforward, can present considerable challenges even for seasoned professionals, and these complexities are often amplified for individuals engaged in smaller-scale projects, such as handymen. Establishing an efficient material procurement process, ensuring the accuracy of material lists, adapting to inevitable project scope changes, securing adequate storage space, maintaining precise inventory counts, and effectively minimizing material waste are all critical aspects that can be time-consuming and demanding. The very act of finding a reliable supplier who can deliver the right materials at the right time and at the right price can be a significant hurdle. Furthermore, unforeseen issues like supply bottlenecks or miscommunications with suppliers can lead to costly delays and disruptions.

For elderly individuals and those with disabilities, these inherent challenges are often compounded by physical limitations. Age-related declines in physical abilities, such as reduced mobility, diminished strength, and impaired vision, can transform once-simple tasks related to material management into significant obstacles. For instance, lifting heavy materials, navigating storage areas, or even reading small print on material labels can become arduous or even dangerous. Similarly, individuals with disabilities may face unique barriers depending on the nature of their impairment, further complicating the already complex process of material management. This confluence of logistical complexities and physical limitations underscores the growing need for technological solutions that can streamline these processes and enhance accessibility for these specific user groups.

1.2 Introduction to Material Minder

The Material Minder mobile application is conceived as a voice-enabled tool specifically designed to address the aforementioned challenges by providing an efficient and accessible platform for managing construction materials for handymen. The application places a particular emphasis on catering to the needs of elderly individuals and those with disabilities, offering features that aim to overcome the limitations they might face with traditional material management methods. At its core, Material Minder provides a comprehensive CRUD (Create, Read, Update, Delete) system, enabling users to easily organize and manage lists of required materials for various construction projects. Recognizing the importance of local context, the application incorporates a customizable database populated with a wide range of construction materials commonly sourced from major Romanian suppliers. This feature not only saves users time in manually

inputting frequently used items but also offers the flexibility to remove irrelevant materials or add custom items unique to their specific needs.

The primary innovation and central focus of Material Minder lies in its integration of a robust voice-enabled input system. This feature is designed to significantly enhance accessibility and ease of use by allowing users to dictate the materials they need, which the application then processes to automatically generate a structured materials list. This hands-free interaction paradigm offers distinct advantages for users who may have limited mobility, visual impairments, or simply prefer a more efficient method of data entry. By combining a user-friendly material tracking system with the power of voice recognition technology, Material Minder aims to provide a solution that directly addresses the inefficiencies and inaccessibility of traditional material management methods, ultimately enhancing productivity and independence for its target users.

2 Theoretical Background

2.1 Human-Computer Interaction (HCI) and Universal Design

Relevance of HCI Principles: Human-Computer Interaction (HCI) is a critical field of study that focuses on the design of technology with the user at the forefront, aiming to create systems that are not only functional but also usable, efficient, and satisfying for the intended audience. The fundamental principles of HCI emphasize the importance of understanding human needs, behaviors, and cognitive processes to develop interactive systems that are intuitive and effective. For Material Minder, a mobile application designed for a diverse user group including elderly and disabled individuals, the application of these principles is paramount.

Key HCI principles such as consistency in design and functionality, providing clear and timely feedback to user actions, ensuring the visibility of important system elements and options, designing for error prevention and offering mechanisms for recovery, promoting learnability through intuitive interfaces, and maintaining simplicity by avoiding unnecessary complexity are all directly relevant to Material Minder’s potential success. Consistency in the placement of navigation elements and the use of terminology, for example, can significantly reduce the cognitive load on users. Providing auditory or visual feedback when a voice command is recognized or a material is added to a list assures the user that their interaction was successful. Making the voice input activation button and the material list clearly visible on the screen enhances usability. Incorporating confirmation dialogues before deleting a material list helps prevent accidental data loss. Designing the application with familiar interaction patterns and clear instructions will improve learnability. Finally, a minimalistic design that focuses on essential features will reduce confusion and enhance efficiency.

Given that Material Minder is intended for use on mobile devices, adherence to mobile-specific HCI considerations is also essential. These include optimizing

touch targets to be of adequate size and spacing for users with potential dexterity issues , simplifying navigation to accommodate one-handed use and the often-interrupted nature of mobile interactions , minimizing the amount of content displayed on a single screen to reduce cognitive overload , and ensuring that the application is responsive and performs efficiently to avoid frustrating users. The principles of matching the user experience and expectations, ensuring functional minimalism, and reducing cognitive load are particularly pertinent for the target demographic.

Insight: The thoughtful application of established HCI principles is not merely a matter of best practice for Material Minder; it is a fundamental requirement for creating an application that is both effective and accessible to its diverse user base, particularly elderly and disabled individuals who may have specific interaction needs and a lower tolerance for poorly designed interfaces.

Application of Universal Design: Universal Design (UD) offers a powerful framework for creating products and environments, including digital applications like Material Minder, that are inherently usable by a wide range of individuals, regardless of their abilities or disabilities, without the need for specialized adaptations. By embracing the seven principles of Universal Design, the development of Material Minder can ensure a high level of inclusivity and accessibility from its inception.

The principle of Equitable Use dictates that the design should be useful and marketable to people with diverse abilities, suggesting that Material Minder’s voice input and customizable database should benefit all handymen, not just those with disabilities. Flexibility in Use implies that the design should accommodate a wide range of individual preferences and abilities, which could be achieved by offering both voice and touch-based input methods, as well as customizable display options. Simple and Intuitive Use is crucial for Material Minder’s target users, emphasizing the need for an interface that is easy to understand and use regardless of experience, knowledge, or concentration level, which can be achieved through clear labeling, consistent navigation, and minimal complexity. Perceptible Information requires that the design communicates necessary information effectively to the user, regardless of ambient conditions or sensory abilities, suggesting the need for visual, auditory, and perhaps even tactile feedback options. Tolerance for Error is particularly important for users who may be prone to mistakes, highlighting the need for features like confirmation prompts for critical actions and easy ways to undo errors. Low Physical Effort means the design can be used efficiently and comfortably with minimal fatigue, which is directly addressed by Material Minder’s voice-enabled input system. Finally, Size and Space for Approach and Use suggests that appropriate size and space should be provided for interaction regardless of the user’s body size, posture, or mobility, implying the need for adequately sized touch targets and a layout that accommodates different holding styles.

Specifically for elderly users, applying UD principles translates to design choices such as large, easily visible UI elements, highly legible and adjustable font sizes, high color contrast between text and background, ample white space to reduce visual clutter, and simple, consistent navigation patterns. For users

with disabilities, UD necessitates the integration of accessibility features from the outset, including ensuring compatibility with screen readers through proper labeling and semantic markup, providing options for adjustable text sizes and color contrast, supporting alternative input methods like voice, and adhering to established web and mobile content accessibility guidelines (WCAG).

Insight: By proactively embedding Universal Design principles into the very fabric of Material Minder’s design, the application can ensure inherent accessibility for a broad spectrum of users, thereby minimizing the need for later retrofitting and maximizing its potential to be a truly inclusive and effective tool for all handymen, regardless of their age or abilities.

Key Takeaway: The successful realization of Material Minder’s potential hinges on a robust theoretical framework grounded in both the general principles of HCI for overall usability and the specific principles of Universal Design to guarantee comprehensive accessibility for its diverse target user group.

2.2 Speech and Language Processing

Fundamentals of SLP and Voice Recognition: The Material Minder thesis proposal rightly emphasizes voice-enabled input as a core innovation to enhance accessibility and ease of use. This highlights the critical role of Speech and Language Processing (SLP) in the application’s design and functionality. SLP is an interdisciplinary field that focuses on enabling computers to process, understand, and generate human language. Voice recognition, a key component of SLP, specifically deals with the conversion of spoken language into a format that computers can understand, typically as text. This process involves several complex steps, including the analysis of the acoustic signal to identify phonemes (the basic units of sound), the use of language models to predict the most likely sequence of words based on context and grammar, and the final transcription of the spoken words into text.

For Material Minder, a fundamental understanding of these processes is crucial for selecting and implementing the most appropriate voice recognition technology. Factors such as the accuracy of the system, its ability to handle different speaking styles and accents (including those prevalent in Romania), its robustness in noisy environments (like construction sites), and its support for the Romanian language will be paramount to the user experience. Furthermore, considering the potential for speech impairments among some elderly or disabled users, the chosen voice recognition system should ideally be adaptable and offer features that can accommodate variations in speech patterns.

Implementation in Mobile Applications: To effectively integrate voice input into Material Minder, it is essential to examine how existing mobile applications utilize this technology for similar purposes, such as inventory management or task organization. Analyzing the user interface design, the methods for initiating voice commands, the feedback mechanisms provided to the user, and the strategies for handling errors or ambiguities in voice input can offer valuable insights and best practices for Material Minder’s development.

Several platforms and APIs are available for incorporating voice recognition

into mobile applications. Operating systems like Android and iOS provide native speech-to-text capabilities that can be leveraged. Additionally, cloud-based services offered by companies like Google, Amazon, and Microsoft often provide more advanced features, including enhanced accuracy, support for a wider range of languages, and the ability to process natural language commands. The selection of the most suitable technology will depend on various factors, including the desired level of accuracy, the need for Romanian language support, the application’s reliance on internet connectivity, the cost of using these services, and considerations around user privacy and data security. The design of the voice interaction within Material Minder should be intuitive and user-friendly. Users need clear visual or auditory cues to indicate when the voice input is active and when it has finished processing. The application should also provide immediate feedback by displaying the transcribed text and allowing users to easily review and correct any errors, either through voice commands or a manual interface.

Insight: The successful implementation of a reliable and user-friendly voice-enabled input system is central to Material Minder’s value proposition, especially for its target demographic. Careful selection of the underlying SLP technology and thoughtful design of the voice interaction will be critical factors in determining the application’s overall effectiveness and user satisfaction.

2.3 Assistive Technologies

Relevance to Material Minder’s Target Users: Given its specific aim to assist elderly individuals and those with disabilities in the often-challenging task of managing construction materials, Material Minder directly aligns with the core principles and goals of assistive technology (AT). The World Health Organization (WHO) defines assistive technology as the application of organized knowledge and skills related to assistive products, including systems and services, where assistive products are any external products designed to maintain or improve an individual’s functioning and independence, thereby promoting their well-being.

For handymen with mobility impairments, Material Minder’s voice-enabled input system offers a crucial alternative to traditional touch-based interfaces, enabling them to manage their material lists and inventory in a hands-free manner. This can significantly enhance their ability to use the application while working or when their hands are otherwise occupied. For individuals with visual impairments, adherence to accessibility guidelines, such as providing high color contrast, ensuring compatibility with screen readers that can audibly convey on-screen information, and allowing for adjustable text sizes, are essential assistive technology considerations. Similarly, for users with cognitive disabilities, a user interface that is simple and intuitive, coupled with clear instructions, consistent navigation, and features that offer tolerance for errors, are vital aspects of assistive design that can make Material Minder more accessible and usable.

Impact on Productivity and Independence: Extensive research indicates that assistive technologies can play a transformative role in improving the productivity and fostering greater independence for individuals with disabilities. By

providing tools and functionalities that help overcome functional limitations, assistive technologies empower individuals to perform tasks more efficiently and with a greater degree of autonomy.

In the context of Material Minder, the integration of voice-enabled input has the potential to directly enhance productivity for handymen who may have physical limitations affecting their ability to use touchscreens or keyboards effectively. The ability to dictate material requirements and manage lists through voice commands can streamline their workflow and allow them to focus on the physical tasks of their projects. Furthermore, a well-organized and readily accessible material management system can contribute to greater independence by enabling elderly and disabled handymen to plan their projects, track their materials, and manage their inventory with less reliance on assistance from others. This sense of control and self-sufficiency can be particularly empowering and can significantly improve their overall work experience.

Insight: By consciously framing Material Minder as an assistive technology, the thesis can draw upon a substantial body of research and established best practices in the design and implementation of accessible technologies for individuals with diverse needs. This perspective not only highlights the application's potential to significantly improve the lives and work of its target users but also provides a valuable framework for guiding its design, development, and evaluation processes.

3 Application Description

The Material Minder application shall provide the following core functionalities:

3.1 Project Management:

- The application shall allow users to view a list of their current projects. Each entry in the list shall display the project name and its creation date.
- The application shall enable users to initiate new projects.
- The application shall provide a manual input method for users to specify the name and date of a new project.
- The application shall allow users to modify existing project details, including the project name and date.
- The application shall enable users to remove projects from their list.

3.2 Material Management:

- The application shall provide users with access to a comprehensive list of construction materials, accessible via a designated "wheelbarrow" icon. This catalog will be populated with data from major Romanian suppliers.

- When creating a new project, the application shall allow users to add construction materials to the initial project list.
- The application shall enable users to add construction materials to existing project lists at any time after the project has been created.
- The application shall allow users to add new construction items to the global catalog and modify the details of existing items (e.g., name, unit, price, supplier).
- The application shall enable users to remove construction items from both individual project lists and the global material catalog (with appropriate confirmation to prevent accidental deletion from the catalog).

3.3 Voice-Enabled Input:

- The application shall feature a clearly identifiable button (indicated by a microphone icon) to initiate voice input for adding materials to a project.
- Upon activation, the application shall utilize voice recognition technology to transcribe the user's spoken input into text.
- The application shall attempt to automatically identify and add recognized construction materials from the global catalog to the current project list based on the transcribed voice input.
- If a spoken material item is not found within the application's database, the system shall create a new construction item in the global catalog. This new item will initially only contain the name as transcribed from the user's voice input. The user shall be prompted or allowed to add further details (e.g., unit, price) to this newly created item later.

3.4 Cost Estimation:

- When a user selects and views a specific project, the application shall automatically calculate and display the total estimated price for all the construction materials currently included in that project list. This calculation will be based on the unit prices stored for each material in the global catalog.

3.5 Non-Functional Requirements

The Material Minder application shall also adhere to the following non-functional requirements:

- **Usability:** The application shall be intuitive and easy to navigate, particularly for elderly individuals and users with disabilities. The user interface should be clear, with appropriately sized elements and sufficient contrast.

- **Accessibility:** The voice-enabled input system shall provide a viable and efficient alternative to manual data entry, significantly improving accessibility for users with motor impairments or visual difficulties.
- **Performance:** The application shall be responsive and perform actions (e.g., loading lists, adding items, calculating costs) within an acceptable timeframe (e.g., under 2 seconds for most operations).
- **Reliability:** The application shall be stable and operate without frequent errors or crashes. Data integrity shall be maintained throughout the application's use.
- **Security:** User data, including project lists and custom material entries, shall be stored securely to prevent unauthorized access.
- **Maintainability:** The application's codebase should be well-structured and documented to facilitate future updates, bug fixes, and the addition of new features.
- **Platform Compatibility:** The application shall be compatible with the target mobile operating system (e.g., Android, iOS).
- **Data Integration:** The application should be designed to allow for potential future integration with other relevant systems or databases.

These requirements provide a comprehensive overview of the functionalities and qualities expected of the Material Minder mobile application, emphasizing its focus on user-friendliness, accessibility, and efficient material management for its target demographic.

3.6 System Specification for Material Minder

This chapter details the specifications for the Material Minder mobile application, building upon the requirements outlined in the previous chapter. It provides a more granular description of the system's components, functionalities, and data handling processes.

4 Functional Specifications

This section elaborates on the functional requirements, providing specific details on how each functionality will be implemented.

4.1 Project Management Specifications:

- **Project List View:** The application will display a scrollable list of projects. Each list item will minimally include the project name (text field, maximum 100 characters) and the project creation date (date field, format YYYY-MM-DD). The list will be ordered by creation date, with the most recent projects appearing first.

- **Project List Interaction:** Users will be able to tap on a project list item to navigate to the detailed view of that project's material list (as specified in Section 4.1.2).
- **New Project Interface:** A dedicated screen or modal will be provided for adding new projects. This interface will include input fields for the project name (text field, maximum 100 characters) and the project date (date picker, defaulting to the current date). A "Save" button will create the new project, and a "Cancel" button will discard the input.
- **Manual Project Creation Input:** The "New Project" interface will present a text input field for the project name and a date picker for the project date. Input validation will ensure that the project name is not empty.
- **Project Modification Interface:** Upon selecting to modify a project, users will be presented with an interface similar to the "New Project" interface, pre-populated with the existing project name and date. Users can modify these fields and save the changes using a "Save" button. A "Cancel" button will discard any modifications.
- **Project Deletion Confirmation:** When a user initiates the deletion of a project, a confirmation dialog will appear, prompting the user to confirm their action before the project and its associated material list are permanently removed.

4.2 Material Management Specifications:

- **Global Material Catalog Display:** Tapping the "wheelbarrow" icon will navigate the user to a screen displaying a comprehensive, searchable list of construction materials. Each item will display the material name. Optionally, unit and default price may also be displayed.
- **Global Material Search:** A search bar will be available on the global material catalog screen, allowing users to filter the list based on keywords in the material name.
- **Adding Materials During Project Creation:** The "New Project" screen will include an option (e.g., a button or section) to add initial materials to the project list. This will likely involve navigating to a subset or filtered view of the global material catalog or providing a voice input option.
- **Adding Materials to Existing Projects:** When viewing a specific project, a dedicated button or option (e.g., "Add Material") will allow users to add new materials to the project's list. This will provide access to the global material catalog (potentially with a filter for commonly used items) and the voice input functionality.

- **New Material Creation Interface:** A dedicated screen or modal will allow users to add new construction items to the global catalog. This interface will include input fields for the material name (text, maximum 100 characters), unit (dropdown or text, maximum 20 characters), and optionally a default price (numeric). A "Save" button will add the new item, and a "Cancel" button will discard the input.
- **Material Modification Interface:** When a user selects to modify a material from the global catalog, an interface similar to the "New Material Creation Interface" will be displayed, pre-populated with the existing material details. Users can modify these fields and save the changes.
- **Material Deletion from Project:** Within a specific project's material list, users will have an option (e.g., a swipe action or a delete button next to each item) to remove a material from that project.
- **Material Deletion from Catalog Confirmation:** When a user initiates the deletion of a material from the global catalog, a confirmation dialog will appear, warning the user that this action will permanently remove the item and asking for confirmation.

4.3 Voice-Enabled Input Specifications:

- **Voice Input Button:** A prominent button with a microphone icon will be located on the screens where adding materials to a project is possible (during creation and in the project detail view).
- **Voice Input Processing:** Upon tapping the microphone button, the application will activate the device's microphone and initiate voice recognition. The recognized speech will be displayed to the user as text for confirmation.
- **Material Matching Algorithm:** The application will employ an algorithm to match the transcribed text against the names of materials in the global catalog. The algorithm should be robust to minor variations in phrasing.
- **Successful Match Handling:** If a match is found, the corresponding material will be added to the current project's material list. The user may receive a confirmation message.
- **Unrecognized Item Handling:** If no exact or sufficiently close match is found in the global catalog for the transcribed text, a new material item will be created in the global catalog with the transcribed text as its name. The user will be informed that a new item has been created and may be prompted to add further details (unit, price) later. This new item will also be added to the current project's material list.

4.4 Cost Estimation Specifications:

- **Project Cost Calculation Logic:** When a project's detail view is accessed, the application will iterate through the materials in the project list. For each material, it will retrieve the associated unit price from the global catalog. The total estimated price for the project will be calculated by summing the prices of all materials in the list.
- **Cost Display:** The total estimated price will be clearly displayed to the user within the project's detail view. The currency will be Romanian Lei (RON).
- **Handling Missing Prices:** If a material in the project list does not have a defined price in the global catalog, it will either be excluded from the cost calculation or indicated to the user (e.g., with a warning message).

4.5 Non-Functional Specifications

This section provides more specific details regarding the non-functional requirements.

- **UI Consistency:** The user interface will adhere to established mobile UI design principles and maintain a consistent look and feel throughout the application.
- **Font Sizes and Contrast:** Text elements will use legible font sizes (minimum 16pt for primary text) and provide sufficient color contrast to ensure readability for users with visual impairments.
- **Voice Input Accuracy:** The voice recognition service should achieve a high level of accuracy in transcribing common construction material names in Romanian.
- **Data Loading Time:** Lists of projects and materials should load within 1 second for datasets of up to 100 items.
- **Action Response Time:** Actions such as adding or deleting items should be reflected in the UI within 0.5 seconds.
- **Data Persistence:** Project data and the global material catalog will be persistently stored on the device (e.g., using a local database).
- **Local Data Security:** Sensitive user data will be protected using appropriate local storage mechanisms and, if necessary, encryption.
- **Code Readability:** The application's codebase will follow established coding standards and include comments to enhance readability and maintainability.

- **API Design for Integration (Future):** The application’s architecture should consider potential future integration with external APIs by adopting a modular design.

These specifications provide a more detailed blueprint for the development of the Material Minder application, outlining the specific behaviors and characteristics of its various components and functionalities.

5 Development and Implementation

This chapter details the development process of the **Material Minder** application, including the technologies used, the implementation of key features, and the overall system architecture. The application was designed to provide an efficient and accessible solution for managing construction materials, with a particular focus on usability for elderly individuals and those with disabilities.

5.1 Technology Stack

The **Material Minder** application was developed using the following technologies:

- **Programming Language:** [Specify the programming language used, e.g., Java, Kotlin, Swift]
- **Framework:** [Specify the framework, e.g., Spring Boot for backend, React Native for mobile]
- **Database:** [Specify the database, e.g., SQLite for local storage]
- **Voice Recognition:** Azure Speech to Text
- **Web Scraping:** [Specify the libraries or tools used, e.g., Jsoup, Selenium]

5.2 Core Functionalities

The application implements the following core functionalities:

CRUD Functionality The application provides a comprehensive CRUD (Create, Read, Update, Delete) system for managing construction materials and projects. This allows users to organize and maintain lists of required materials for various construction projects. The controllers `ConstructionItemController.java`, `ProjectController.java`, and `UserController.java` provide the API endpoints for these operations. For example, the `ConstructionItemController.java` defines endpoints for retrieving, creating, updating, and deleting construction items.

Voice-Enabled Input A key feature of Material Minder is the integration of voice-enabled input using Azure Speech to Text. Users can dictate the materials they need, and the application processes the input to automatically generate a structured materials list. The process involves the following steps:

1. The user speaks into the device's microphone.
2. The audio is captured and sent to the Azure Speech to Text service.
3. Azure Speech to Text converts the audio into text.
4. The text is processed by the application to identify the requested materials.
5. The materials are added to the user's project.

Data Acquisition To provide users with a comprehensive database of construction materials, data is scraped from major Romanian suppliers, Dedeman and Leroy Merlin. This process involves:

- Identifying the target websites and their structure.
- Using web scraping tools to extract relevant data, such as material names, descriptions, and prices.
- Cleaning and formatting the extracted data.
- Storing the data in the application's database.

5.3 System Architecture

[Include a diagram or description of the system architecture. For example:]

- A multi-layered architecture consisting of a presentation layer (mobile app), a business logic layer (backend), and a data access layer (database).
- A description of how the different components interact with each other.
- Details about API endpoints and data flow.

5.4 Database Design

[Include a description of the database schema. For example:]

- A description of the tables used to store data, such as users, projects, materials, and project items.
- An Entity-Relationship Diagram (ERD) to illustrate the relationships between the tables.
- Details about the data types and constraints used for each field.

5.5 User Interface (UI) Design

[Include a description of the user interface design, focusing on accessibility and usability. For example:]

- Large, clearly visible UI elements.
- High color contrast between text and background.
- Simple and consistent navigation patterns.
- Adequately sized touch targets.
- Support for both voice and touch-based input.

5.6 Implementation Details

[Provide more specific details about the implementation of key features. For example:]

- Code snippets illustrating how the voice input is handled.
- Details about the web scraping process, including the tools and techniques used.
- Explanation of how CRUD operations are implemented for different entities.
- Error handling and validation mechanisms.
- Security considerations, such as data protection and authentication.

5.7 Testing

[Describe the testing process, including the types of tests performed and the tools used. For example:]

- Unit tests for individual components and functions.
- Integration tests for verifying the interaction between different components.
- User testing with target users (elderly and disabled individuals) to evaluate usability and accessibility.
- Performance testing to ensure the application meets the non-functional requirements.

- 6 Results and Evaluation
- 7 Discussion
- 8 Conclusion
- 9 References